Database concepts

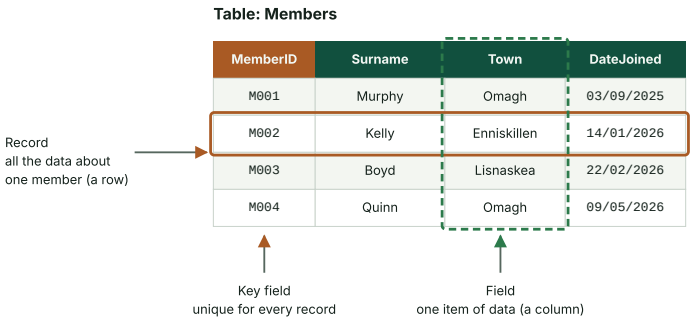
L08
FACT FILE

Specification: students should be able to

- Demonstrate understanding of and explain basic database concepts such as table, record, field, key field, query, form, report, macro, relationship and importing data
- Explain the difference between a flat file and a relational database
- Explain how a primary key and a foreign key are used to create a relationship between tables
- Describe one-to-one, one-to-many and many-to-many relationships and the purpose of referential integrity
- Describe the purpose of queries, forms, reports and macros in a database application

CCEA fact file: Fact File 07: Database Applications (1)

Key words: database, table, flat file, record, field, key field, primary key, composite key, foreign key, relationship, one-to-many, referential integrity, query, form, report, macro, importing, data redundancy



A database table showing a record, a field and the key field

What a database is and why we use one

A database is an organised collection of related data, stored so that it can be searched, sorted and presented quickly. Almost every organisation in Northern Ireland uses one: schools keep pupil details and attendance, GP surgeries keep patient records, shops keep stock levels and sports clubs keep member lists. Inside the database, data is held in tables. A table stores data about one type of thing, for example members, and is laid out in rows and columns. A database package also gives tools to work with the data. Queries find the records you need, forms make it easy to enter and edit data, reports present the results neatly, and macros automate jobs that are done again and again. Tables can be linked by relationships so that each fact only needs to be stored once. In this lesson you will learn the key terms that the exam expects you to define and to spot in a table.

Records, fields and key fields

Each row in a table is a record: all of the data about one person or item, such as one swimmer in a club. Each column is a field: a single item of data that is stored for every record, such as Surname or DateOfBirth. The column heading is the field name, and each field is given a data type. The key field, also called the primary key, is the field that uniquely identifies each record. No two records can have the same key value and it cannot be left blank. Codes such as MemberID or PupilID make good key fields because the organisation creates them to be unique. Names, dates of birth, phone numbers and towns make poor key fields because two people can share them. Sometimes no single field is unique, so two fields are used together; this is called a composite key, for example MemberID and SessionDate in a table of lesson attendance.

Flat files, relationships and referential integrity

A flat file keeps all the data in one table. This often means the same details are typed many times, which is called data redundancy. It wastes space and causes errors, because one copy may be updated while another is not. A relational database avoids this by splitting the data into several tables, each about one thing, and linking them. The link is made by storing the primary key of one table in the other table, where it is called a foreign key. For example, MemberID is the primary key of Members and a foreign key in Lessons. This is a one-to-many relationship: one member can book many lessons. Other types are one-to-one and many-to-many. Enforcing referential integrity on the relationship keeps the tables consistent. It stops a lesson being booked for a member who does not exist, and stops a member being deleted while bookings still point to them, unless cascade delete is switched on.

Queries, forms and reports

Tables store the data, but users rarely work with tables directly. A query is a question asked of the database. It checks every record and returns only those that meet the criteria, for example all Junior squad members born after 2011. A query can use fields from more than one linked table and can sort the results. A form is an on-screen layout used to enter new records, edit existing ones or view records, usually one at a time. Forms are easier to use than a table, can include drop-down lists, buttons and the club logo, and can combine fields from more than one table while saving each item to the correct table. They also help keep data accurate by limiting what can be typed. A report presents data from a table or query in a formatted layout ready for printing or display. Reports can group similar records, sort them and calculate totals or counts, for example a register of members grouped by squad.

Macros and importing data

A macro is a small stored program that performs a set of database actions automatically. Instead of carrying out several steps every time, the user clicks one button on a form and the macro does the work. Macros can open a form, run a query, print a report, or add, update or delete a record. These four record jobs are sometimes called CRUD: create, read, update and delete. Macros save time, reduce mistakes and let staff with little database knowledge use the system safely. Importing data means bringing data into the database from another file rather than typing it again. A common example is importing a csv file or a spreadsheet worksheet, such as a list of new members exported from an online sign-up form. Importing is quicker and avoids typing errors. Before importing, the data should be checked so that column headings match the field names, values suit each field's data type and there are no duplicate key values.

T Key terms from the mark schemes

Database
An organised, structured collection of related data that can be searched, sorted and presented, usually held in one or more tables.

Flat file
A database that holds all its data in a single two-dimensional table, which often leads to repeated data.

Field
A single item of data stored about every object or person, forming a column in a table and identified by its field name.
`Credited: U1 2019 Q3, 2021 Q2, 2024 Q3`

Table
A database structure that stores data about one type of thing, organised in rows (records) and columns (fields).
`Credited: U1 2019 Q3, 2024 Q3`

Record
All of the related data about one object or person, forming a complete row in a table. A record is made up of fields.
`Credited: U1 2019 Q3, 2023 Q3, 2024 Q3`

Key field / primary key
A field whose value is unique for every record, used to identify each record in a table.
`Credited: U1 2019 Q3, 2022 Q10, 2024 Q3`

Composite key
A primary key made up of two or more fields that together uniquely identify a record.

Foreign key
The primary key of one table stored as a field in a second table, used to link the two tables.

Relationship
A link between two tables made through a common field (primary key and foreign key). It can be one-to-one, one-to-many or many-to-many.

Relational database
A database in which data is held in several linked tables.
`Credited: U1 2021 Q2, 2024 Q3`

One-to-many relationship
One record in a table links to many records in another table, e.g. one customer makes many bookings.
`Credited: U1 2019 Q3, 2021 Q2`

Cardinality
How the records in two related tables relate to each other: one-to-one, one-to-many or many-to-many.

Crow's foot
A symbol drawn at the 'many' end of a relationship line, used instead of the infinity sign (∞).

Referential integrity
A rule applied to a relationship that keeps the linked tables consistent, e.g. a foreign key value must match an existing primary key.

Form
An on-screen layout used to enter new records, edit existing records or view records in a database.
`Credited: U1 2022 Q10, 2023 Q3`

Report
A database tool that presents data from a table or query in a formatted, user-friendly layout for printing or display, often grouped and sorted.
`Credited: U1 2019 Q3, 2022 Q10, 2023 Q3`

Macro
A small stored program that carries out a repetitive task automatically, usually at the click of a button.
`Credited: U1 2021 Q3, 2022 Q10, 2024 Q10`

Import
To bring data into an application from another file, such as a csv file or spreadsheet, instead of retyping it.
`Credited: U1 2022 Q10`

Exam focus

Database questions appear on nearly every Unit 1 paper, often worth 8 to 11 marks: 2019 Q3, 2021 Q2, 2022 Q10, 2023 Q3, 2024 Q3 and 2025 Q5. You are often given a table and asked to name a record, a field or the key field, or to fill in missing field names. When asked to choose a key field, name it and justify it: the value is unique for each record. Say why other fields are unsuitable, for example two customers could have the same surname. For relationship questions (2019 Q3, 2021 Q2) name the type, usually one-to-many, and say which table is the 'one' end. Questions on database tools (2022 Q10, 2023 Q3) and key term matching (2024 Q10) test whether you can tell a query, form, report and macro apart, so learn a clear one-sentence definition of each. A common mistake is saying a report searches for data; it is the query that searches and the report that presents the results.

`Unit 1 2025 Q5`	`[11]`	Data types, validation checks, data processing
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q3`	`[11]`	Field names and data types, database terms, key field, query design
`Unit 1 2023 Q3`	`[9]`	Database concepts; validation rules
`Unit 1 2022 Q10`	`[11]`	Database tools, field types, importing csv, validation

? Check yourself

1. Define the terms (a) record and (b) field. [2]
2. Look at the Erne Valley Swimming Club Members table on the slides. (a) State how many records the table contains. (b) Identify the key field and justify your choice. (c) Explain why Surname would not be a suitable key field. [4]
3. Describe the purpose of a form in a database, and give two ways a form can help to keep the data accurate. [3]

Answers: 1. (a) All the data about one person or item / one row in a table (1). (b) One item of data stored about each record / one column in a table (1). 2. (a) 4 (1). (b) MemberID (1) because each member has a different/unique ID (1). (c) Two members can have the same surname, e.g. Maguire (1) so it would not uniquely identify a record (1). 3. Purpose: an on-screen layout used to enter new records / edit or view existing records (1). Any two from: drop-down lists limit choices (1) / validation rules on fields reject unsuitable data (1) / input masks control format (1) / only the fields needed are shown so fewer mistakes (1) / data is saved to the correct table automatically (1).